Buzz: Build and Change Report

AI Reel Studio: reels from photos, videos or just an idea | version 1.0.0 | 12 Nov 2026

Buzz turns a stack of photos and videos, or just a written idea, into a ready-to-post 9:16 Instagram reel with music, text, a caption and five hashtags. It runs on a machine you control and works on its own in the background.

At a glance

Size	26 Python files, about 4,385 lines
Runs on	Windows 10 or 11, CPU only
Engine	Python 3.12, FFmpeg, librosa, OpenCV, Streamlit; Claude for understanding and planning
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What it does

Two ways to start	Upload photos and videos (with your own song, music made for you, or silent), or type an idea and get a finished reel from the words alone.
Smart editing	The AI studies your pictures and clips, plans a story, cuts on the beat, keeps text clear of faces, and lets you choose the opening and closing shot.
Instagram pack	Hook, caption, 5 hashtags, cover text, best time to post and alt text for every reel.
Idea reels	Ten colour themes, with soft gradient, plain colour, drawn pictures or stock video behind the words.
Voiceover	A free voice that runs on your PC (Kokoro), or Google Cloud Text-to-Speech with your own key. Music dips under the voice.
Your control	Reel length from 15 to 90 seconds, how many pictures to use (Auto, All or a number), change notes in plain words, optional logo.
Looks after itself	Live progress, a background worker that restarts itself, accounts with sign-in and remember-me, safe uploads, and a nightly clean-up of old reels.

What you need

- Windows 10 or 11 with a normal user account (no administrator needed), about 4 GB of free disk, and internet access.
- A Claude API key from console.anthropic.com (required). Buzz is powered by Claude.
- Optional: a Google Cloud Text-to-Speech key (voiceover with Indian, UK and US voices), a Pexels key (stock video for idea reels), a mailbox for emailing password-reset codes.

Change history

1.0.0 | 12 Nov 2026 | First public release

- Two modes in one app: reels from photos and videos, and reels from only an idea.
- AI understanding of every picture and clip, beat-matched cutting, face-safe text, collages for landscape pictures, optional portrait centring.
- Choose the opening and closing picture or clip; the story is built around them.
- Music: use your own song, or generate original music from up to three keywords, tuned to be clear on phone speakers.
- Voiceover: free local voice (Kokoro) or Google Cloud Text-to-Speech; background music ducks under the voice.

- Idea reels: ten colour themes; gradient, plain colour, drawn pictures or Pexels video backgrounds.
- Instagram pack: hook, caption, 5 hashtags, cover text, best time, alt text; download as text; readable file names.
- Reel length selector (15 to 90 s) and an assets selector (Auto, All, or a number).
- Change notes in plain words ('make shot 5 the opening', 'use a different font') and one-tap quick changes.
- Accounts with email and password, 30-day remember-me, forgot-password codes, owner and member roles.
- Settings: Claude key, free voice install, Google voice key, Pexels key, mail, system status, automatic cleanup.
- Safe uploads: file-type and content checks, size limits, no executable files.
- Live page: auto-refreshing lists and a smooth real-time progress bar.
- Automatic cleanup: reels older than 15 days move to a holding folder and are deleted 3 days later; Keep switch per reel.
- My reels shows the last 15 days.
- Video encoding uses an NVIDIA, Intel or AMD graphics card automatically when one is present and its quality at least matches the CPU encoder; otherwise the CPU.
- Windows installer (no administrator needed) that sets up Python, FFmpeg, the environment and two background tasks.

Modules

worker.py	Background worker: one job at a time, restarts itself
pipeline.py	One folder of media to one finished reel
ingest.py	Reads and catalogues photos and videos
music.py / music_provider.py	Beat and energy analysis; original music generator
ai_plan.py	AI understanding, story plan, asset count, opening and closing shot
render.py / encoder.py	FFmpeg rendering, text, logo, audio mix; picks the CPU or a graphics-card video encoder by testing quality
layout.py / facedetect.py	Keeps text clear of faces
revise.py	Plain-language change notes
prompt_reel.py / art_provider.py	Idea reels: script, scenes, themes, drawn pictures
voice_provider.py / narration.py	Kokoro and Google voices, voiceover timing
stock_provider.py	Pexels stock video
social.py	Instagram pack
progress.py	Real-time progress
uploads.py	Safe upload checks
auth.py / mailer.py	Accounts, sessions, reset codes, optional email
cleanup.py	Nightly tidy-up of old reels
app.py	The web page

Keys used

anthropic	Claude key (required)
google_tts	Google Text-to-Speech key (optional)
pexels	Pexels key (optional)
smtp	Mail settings for reset codes (optional)

Buzz is powered by Claude. The code is open: you are free to change it and plug in the AI of your choice, whichever works for you.